Forces on two objects

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify an equal-and-opposite interaction pair on different objects.

SUCCESS CRITERIA - I can...

1. Name both objects in an interaction pair.
2. Place the two equal-and-opposite forces on different objects.
3. Explain why equal forces can produce unequal accelerations.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

With a fixed net force, a 4 kg system accelerates at 12 m/s². Predict acceleration for a 8 kg system. State a control needed for this comparison.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F_{A \text{ on } B} = -F_{B \text{ on } A}$.
The two forces act on different objects at the same instant. Their accelerations can differ.

Worked reasoning: Two skaters push apart. Each feels 20 N from the other. If their masses are 40 and 80 kg, accelerations are 0.5 and 0.25 m/s² in opposite directions.

Your turn: Draw separate force diagrams for the horizontal interaction. Explain why equal forces do not imply equal accelerations.



Skater A: 40 kg



Skater B: 80 kg

Marzano check - circle: ① ② ③

03 Name both forces

A person presses a wall to the right. On the two-object model, draw the force the hand exerts on the wall and the force the wall exerts on the hand. Label both forces with their source and receiving object.



Hand

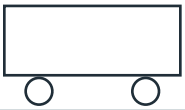


Wall

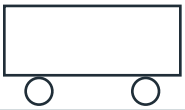
Marzano check - circle: ① ② ③

04 Heavy and light carts

A heavy cart and a light cart push apart on a low-friction track. Add their interaction-force arrows to the model. Compare force sizes and explain which cart accelerates more.



Light cart



Heavy cart

Marzano check - circle: ① ② ③

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05 Balanced or a pair?

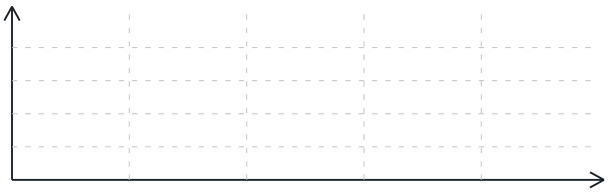
A book rests on a table. Earth pulls book down; table pushes book up. Are these the two forces of one interaction pair? Explain by naming the object receiving each.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Start at 6 m, move to 16 m, then finish at 8 m. Draw the route. Find distance, displacement with direction and final position.
Label the model with the given positions and any prediction.

Route



Position (m)

Marzano check - circle: 1 2 3

07 Exit check

A runner's foot pushes backward on the ground without slipping. Add the two interaction-force arrows to the model. Name the partner force and the object it acts on. Explain how the runner can accelerate forward.



Foot



Ground

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Choose two masses that would give a 3: 1 acceleration ratio while the forces remain equal. Explain which skater accelerates more.

Skater	Mass (kg)	Acceleration ratio
A	Choose: ____	3
B	Choose: ____	1

Marzano check - circle: 1 2 3